

Permutations



1

a 120

e 1

b 42

f 11

c 120

g 210

d 120

h 20 160

2 362 880

3 120

4 120

Probabilities with permutations

5 $\frac{1}{2}$

6

a 24

b 6

c $\frac{1}{4}$

7

a $\frac{5}{6}$

b 72

8

a $\frac{3}{28}$

b $\frac{15}{28}$

9

a 120

b $\frac{1}{5}$

c $\frac{1}{120}$

10

a 40 320

b $\frac{1}{40\,320}$

11

a $\frac{1}{3}$
e $\frac{4}{5}$

b $\frac{2}{3}$



c 0

d $\frac{1}{5}$

12

a $\frac{1}{120}$
e $\frac{3}{5}$

b $\frac{1}{120}$
f $\frac{3}{5}$

c $\frac{1}{120}$

d $\frac{2}{5}$

Applications

13 120

14 5040

15

a $\frac{1}{120}$

b $\frac{1}{120}$

c $\frac{1}{120}$

d $\frac{1}{5}$

16

a 17 576 000

b $\frac{1}{17\,576\,000}$

c $\frac{1}{13}$

17

a 720

b 120

c $\frac{1}{6}$

18

a $\frac{1}{90}$

b $\frac{1}{90}$

c $\frac{1}{45}$

d $\frac{4}{9}$

19

$\frac{1}{2\,793\,510\,720}$

20 $\frac{7}{11}$

21

a 1

b 29 889



22
a $\frac{1}{210}$

b $\frac{1}{35}$

23
a $\frac{1}{156}$

b $\frac{1}{1716}$

24 $\frac{1}{720}$

25 $\frac{1}{22}$